

SEAT No. _____

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SARDAR PATEL UNIVERSITY

External Examination-2020

M. Sc. – CA&IT- 3rd Semester (Regular)

PS03CIIT33 – Compiler Design

30th December, 2020 (Wednesday)

Time: 10:00 am to 12:00 pm

Max. Marks – 70

Q-1 Give answers of following multiple choice questions.

[10]

- 1 The analysis part is often called the _____ of the compiler.
A) Front end B) Back end C) Middle end D) None of these
- 2 What is the output of lexical analyzer?
A) A parse tree B) A list of tokens C) Intermediate code D) Machine code
- 3 Regular expression $(x/y)(x/y)$ denotes the set _____.
A) $\{xy, xy\}$ B) $\{xx, xy, yx, yy\}$ C) $\{x, y\}$ D) $\{x, y, xy\}$
- 4 In regular expressions, the operator '*' stands for _____.
A) Concatenation B) Selection C) Iteration D) Addition
- 5 Grammar of the programming is checked at _____ phase of compiler.
A) Semantic analysis C) Syntax analysis
B) Code optimization D) Code generation
- 6 A grammar that produces more than one parse tree for same sentence is called as _____.
A) Ambiguous B) Unambiguous C) Regular D) All of these
- 7 Which of the following field is not in triple?
A) Op B) arg1 C) arg2 D) result
- 8 The final phase of compiler model is _____.
A) Syntax Analysis C) Code Generation
B) Code optimization D) Lexical Analysis
- 9 An important modification that decreases the amount of code in a loop is _____.
A) Code Motion C) Dead-code Elimination
B) Copy Propagation D) None of these
- 10 Compiler translates the source code to _____.
A) Executable Code C) Machine Code
B) Binary Code D) Both B and C

[P.T.O.]

Q-2 Fill in the blanks and True – False.

[08]

- 1 The synthesis part is often called the _____ of the compiler.
- 2 A compiler program written in a high level language is called _____.
- 3 Syntax Analyser is also known as _____.
- 4 _____ data structure in a compiler is used for managing information about variables and their attributes.
- 5 Lexical analysis is about breaking a sequence of characters into lexemes. (True / False)
- 6 Regular expressions can be used only for values of type string and number. (True / False)
- 7 Recursive descent parser is a top-down parser. (True / False)
- 8 The input of code generator is Source Code. (True / False)

Q-3 Write down the answer of the following short question. (Any Ten)

[20]

- 1 List all the language processors and explain any one of them.
- 2 What is syntax tree?
- 3 What is compiler?
- 4 What is lexical analysis?
- 5 What is transition diagram? Why it is used?
- 6 Define: Lexeme, Token and Pattern
- 7 What is Parser?
- 8 List types of Parser.
- 9 List common programming errors and explain any one of them.
- 10 What is Code Generator?
- 11 Define : Quadruples, Triples, Indirect triples
- 12 What are the principal sources of optimization?

Q-4 Give the answers of following questions. (Any 4)

[32]

- 1 Explain phases of compiler in detail.
- 2 What is language processor? Explain all the language processors in detail.
- 3 Explain input buffering in detail.
- 4 Explain the role of lexical analyzer.
- 5 Explain Top Down Parsing and Bottom Up Parsing.
- 6 Write a detail note on role of the Parser.
- 7 Write a detail note on Issues in the Design of a Code Generator.
- 8 Write a detail note on Symbol Table.